# Modeling of kilonova light curves from neutron star mergers and possible constraints on nuclear physics

by

Diego Fernando Hernández Porras

Submitted to the Department of Physics
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## ABSTRACT

Neutron stars are compact objects whose hydrostatic equilibrium is determined by the equation of state (EOS) of the dense matter in their interiors, which makes them natural laboratories for nuclear physics at densities inaccessible to terrestrial experiments. When two neutron stars merge, part of their mass is ejected and, as demonstrated by the gravitational-wave event GW170817 and its associated kilonova, this ejected material is a site of rapid neutron capture (r-process) nucleosynthesis. The yield of heavy nuclei in the ejecta depends on the properties of neutron-rich nuclei far from stability, including nuclear masses, decay rates, neutron-capture rates, and fission properties, among others, many of which are unknown. Since nuclear physics plays a crucial role in the evolution of observables, understanding the impact of relying on theoretical nuclear physics results is essential to correctly interpret current and future multimessenger observations.

This work presents a detailed analysis of the impact of different theoretical $\alpha$- and $\beta^-$-decay rates on the kilonova evolution. The nucleosynthesis was calculated using the nuclear reaction network WinNet across an ensemble of initial thermodynamic conditions parametrized by the initial electron fraction ($Y_{e,0}$), the expansion timescale ($\tau$) and the initial entropy per baryon ($s_0$). The kilonova light curves were calculated using current semi-analytical models based on radiative-transfer simulations. The results show that the heating rate from radioactive decays deviates from the widely accepted power-law behavior depending on the $\beta^-$-decay prescription consistently across the initial conditions studied. In addition, the set of nuclei responsible for such behavior was identified and the description of their decay chains is presented.

This next paragraph may change as results from GR simulations are being obtained
The results on the impact of theoretical rates described above are then tested in a merger simulation where the resulting kilonova is now described as a superposition of different ejecta components. The amount of ejected mass and its velocity are extracted from a general-relativistic simulation which was computed using the Einstein Toolkit framework.

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# Acknowledgments

Este trabajo respresenta el final de una de las etapas mas bonitas de mi vida. Llevaré con mucho cariño el recuerdo de todos mis amigos a donde me lleve la motivación.

# Contents


**1 Introduction** **6**
- 1.1 Kilonovae in a nutshell . . . 6
- 1.2 A brief history of the kilonova . . . 7
- 1.3 Modern models of kilonovae and observations . . . 8
- 1.4 Objectives and thesis structure . . . 10

**2 From Neutron stars to Kilonovas** **11**
- 2.1 The Neutron star . . . 11
- 2.2 Formation of binary neutron stars . . . 12
- 2.3 Gravitational-wave-driven inspiral and the maximum separation . . . 14
- 2.4 Mass ejection: tidal, shock-driven, and disk-wind ejecta . . . 15
- 2.5 Weak interactions and the electron fraction of the ejecta . . . 15
- 2.6 Electromagnetic counterparts . . . 15

**3 Numerical Relativistic Hydrodynamics** **16**
- 3.1 3+1 decomposition and evolution formulation . . . 16
- 3.2 General-relativistic hydrodynamics in conservative form . . . 16
- 3.3 The Einstein Toolkit: initial data, mesh refinement, atmosphere . . . 16
- 3.4 Gravitational-wave extraction . . . 16
- 3.5 Ejecta diagnostics and unbound-matter criteria . . . 16

**4 r-Process Nucleosynthesis** **17**
- 4.1 Nuclear statistical equilibrium and freeze-out . . . 17
- 4.2 The r-process: neutron-to-seed ratio, fission cycling . . . 17
- 4.3 Reaction-network equations and numerical solution . . . 17
- 4.4 Nuclear inputs: masses, decay rates, neutron capture, fission . . . 17
- 4.5 Radioactive heating and thermalization . . . 17

**5 Kilonova Light Curves** **18**
- 5.1 Li-Paczyński model . . . 18
- 5.2 Radiative transfer and the diffusion regime . . . 18
- 5.3 Opacities of r-process matter . . . 18
- 5.4 Light-curve modeling . . . 18

**6 Methodology on the impact of theoretical decay rates across thermodynamics trajectories** **19**
6.1 Binary neutron-star simulation . . . 19
6.2 From simulation to trajectories: ejecta extraction and expansion . . . 19
6.3 Initial termodinamic trajectories treatment . . . 19
6.4 Nuclear reaction network setup and theoretical reaction reates . . . 19
6.5 Modified Li-Paczyński model and radiative transfer simulations . . . 19

**7 Results** **20**
7.1 Merger dynamics . . . 20
7.2 Dynamical-ejecta properties . . . 20
7.3 Nucleosynthesis and heating rates . . . 20
7.4 Kilonova light curves . . . 20
7.5 Relevant nuclei and uncertainties . . . 20
7.6 Detectability with current and future telescopes . . . 20

**8 Discussion** **21**
8.1 Comparison with previous studies . . . 21
8.2 Limitations and model assumptions . . . 21

**9 Conclusions and Outlook** **22**

**A Quadrupole and $L_{GW}$** **23**

**B Convergence tests** **25**

**C Network and equation-of-state configuration details** **26**

*References* 27

# Chapter 1

# Introduction

Binary neutron star mergers (BNSMs) are among the most energetic events in the Universe, releasing tremendous amounts of energy through gravitational waves, neutrinos, and electromagnetic radiation. During and after the merger of two neutron stars, a fraction of their mass is ejected through tidal interactions, shocks, and disk winds. The ejected material is expected to consist of extremely neutron-rich matter, within which rapid neutron captures (r-process) lead to the formation of heavy nuclei far from stability. Consequently, these mergers are considered one of the primary sites for the synthesis of the heaviest elements in the Universe [1,2]. As the freshly synthesized nuclei decay toward stability, their radioactive energy release drives the thermodynamic evolution of the ejecta and produces a thermal electromagnetic transient [3]. Such a transient is called a kilonova and will be the main topic of this thesis, since it encodes crucial information about the physics of the merger, such as the abundance pattern of the elements and the amount of ejected material.

## 1.1 Kilonovae in a nutshell

Kilonovae are closely related to the nuclear and atomic phenomena inside the ejecta. As heavy nuclei decay toward stability, part of the energy is released in the form of high energy photons according to the specific transitions inside the nuclei. In addition, decay products such as $\alpha$ and $\beta$ particles, which also carry part of the decay energy, interact with surrounding charged particles primarily through Coulomb interaction, producing photons that carry away part of that energy. Effectively, the relevant particles will be photons.

At early stages the density is still high, so photons cannot easily escape and interact with several charged particles before reaching the surface. As photons interact with matter, their energy spectrum approaches that of a blackbody, this process is called thermalization, and the same process applies to all the other decay products [4]. The emission at such stages is well described as a blackbody with a temperature evolution governed by the heating rate from radioactive decays. As the density decreases, bound states such as atoms form in a process called recombination. After this point the thermalization of the photons depends on the emission and absorption spectra of heavy elements. As a consequence, kilonova light curves encapsulate information about the abundance of certain nuclei with specific emission and absorption lines. For example, the presence of lanthanides increases the opacity of the

ejected material by an order of magnitude, so a kilonova with lanthanides will thermalize more efficiently and develops a long-lived thermal spectrum [5]. Since the abundances of lanthanides and of the rest of nuclei are determined by nuclear physics inputs, such as reaction rates, decay rates and masses, a detailed treatment of their uncertainties is needed for the correct interpretation of current and future observations. This is where this research work sits, as developed later in the text.

In the following sections, we will first give the reader a brief historical introduction ot the kilonova. Then we will present modern kilonova models and their limitations. By the end of this chapter, we will state the objective of the research, explain the methodology, and describe the structure of this thesis.

## 1.2 A brief history of the kilonova

Until the last decade, the astrophysical origin of roughly half of the nuclei heavier than iron remained one of the major open problems of nuclear astrophysics. Big Bang nucleosynthesis accounts for the primordial abundances of the lightest nuclides (D, $^3$He, $^4$He and $^7$Li), while hydrostatic and explosive burning in stars and supernovae produces the elements from carbon up to the iron peak and some heavier elements [6,7]. The issue was that, beyond iron, the binding energy per nucleon decreases and Coulomb barriers grow, so heavy nuclei are built mainly by neutron captures rather than by charged particle fusion. Two main channels were known [8]. The first channel is the slow (s) process, in which neutron captures are slower than $\beta$-decays. The second is the rapid (r) process, in which captures are much faster than $\beta$-decays and drive matter far to the neutron-rich side of stability. The s-process is now well understood and operates mainly in asymptotic giant branch (AGB) stars and, for the lighter heavy elements, in massive stars [9]. The conditions required by the r-process have also been characterized to neutron densities $n_n \gtrsim 10^{20}\,\mathrm{cm}^{-3}$, or equivalently suitable combinations of electron fraction, entropy and expansion timescale in the ejecta [10]. However, the site where these conditions occur remained elusive.

For several decades the favored candidate was the core-collapse supernova, in particular the neutrino-driven wind from the newborn neutron star [11]. Modern simulations, however, find this wind neither neutron-rich nor entropic enough to produce the heaviest $r$-process nuclei [12], so ordinary supernovae contribute at most a weak $r$-process, producing elements up to about Sr–Ag. Observations point in the same direction. The large star-to-star scatter of europium in metal-poor stars [13], the strong $r$-process enrichment of the ultra-faint dwarf galaxy Reticulum II [14], and the low abundance of live $^{244}$Pu in deep-sea sediments [15] all indicate that the heavy $r$-process elements are made in rare events with large yields, rather than in frequent ordinary supernovae.

Neutron star mergers, on the other hand, match those requirements. Early theoretical studies had proposed that during the merger of compact objects, tidal forces lead to the ejection of neutron-rich material [16]. These ejecta were expected to be very neutron-rich and decompress rapidly, ideal for r-process and enable the synthesis of heavy neutron-rich nuclei. It was then proposed that the radioactive decay of freshly synthesized nuclei would power a short-lived optical transient. In 1998, Li and Paczyński presented a simple model (LP98) to calculate that possible electromagnetic counterpart. The model predicted a light

curve with a bolometric luminosity peak of approximately $10^{44}\ erg\ s^{-1}$ at $t \approx 1$ day after the merger depending on free parameters such as the expansion velocity, the ejected mass and an effective constant for the radioactive heating [17].

## 1.3 Modern models of kilonovae and observations

The LP98 model assumes an analytical energy generation rate of the form $\dot{\varepsilon}(t) \propto t^{-1}$ and models the electromagnetic emission as a blackbody spectrum. However, these assumptions may not fully capture the physical complexity of the event. As we discuss before the $\gamma$-rays produced by nuclear decays can escape without thermalizing, leading to deviations in the predicted emission spectrum. [As we discuss before the → As discussed before, the.] In addition, the actual energy generation rate may deviate from the assumed analytical form, making a more detailed treatment of the spectral evolution of the emission a challenging task.

The first attempt to construct a realistic model of this phenomenon was presented by Metzger et al. [3], who performed self-consistent calculations of the optical and near-infrared counterparts with two major improvements. First, they implemented nuclear reaction network calculations to evolve the abundances and compute the corresponding energy generation rate, $\dot{\varepsilon}(t)$, using both experimental and theoretical reaction and decay rates. Second, they performed one-dimensional Monte Carlo radiative transfer simulations, assuming opacities derived from bound–bound transitions of iron (Fe), with corrections applied to the ionization energies, in order to model the resulting light curves. The first results showed an abundance pattern close to the solar system r-process abundances. Also, the energy generation rate was found to follow a power-law dependence, $\dot{\varepsilon}(t) \propto t^{-\alpha}$ with $\alpha \sim 1.1 - -1.3$, and remarkably robust against variations in the electron fraction ($Y_e \sim 0.05 - -0.35$), thermodynamic trajectories, and nuclear properties along the r-process path. The second key result demonstrated that the light curves evolve significantly differently from those predicted by the simple blackbody approximation. The simulated light curves exhibited a characteristic "bump-like" behavior on different timescales across various wavelength bands and a peak luminosity three orders of magnitude greater than that of a typical nova, the latter being the reason why the term kilonova was first introduced.

However, radiative transfer simulations face major challenges due to the highly complex opacities arising from the dense line spectra of heavy r-process nuclei. This "opacity problem" has been addressed through several approaches. For instance, in Ref. [18], atomic structure models for representative elements were employed to compute radiative transition rates for tens of millions of lines, enabling more realistic estimates of the opacities for r-process material. A different approach was taken in Ref. [19] where the opacity was estimated by taking into account electron scattering, free–free, bound–free and bound–bound transitions using analytical formulas. These advances led to the identification of distinct kilonova components based on the lanthanide content of the ejecta [20,21]. Material with a low electron fraction ($Y_e \leq 0.22$) is expected to produce lanthanide-rich ejecta with high opacity, resulting in a red, longer-lived transient commonly referred to as the red kilonova. In contrast, material ejected with a higher electron fraction ($Y_e \geq 0.3$), typically from the polar regions, evolves into lanthanide-poor ejecta with lower opacity, producing a short-lived, blue optical transient known as the blue kilonova.

Nevertheless, all previous efforts remained purely theoretical and computational, as no observational data were available to validate the models. This situation changed on August 17, 2017, when the Advanced LIGO and Advanced Virgo gravitational-wave detectors made their first observation of a binary neutron star merger (BNSM) [22] with the subsequent identification of the electromagnetic counterparts in the same location, which supported the identification of the event as a BNSM. An approximately $100\,s$ long gravitational-wave signal, GW170817, was followed by a short gamma ray burst (sGRB), GRB 170817A, and an optical transient named SSS17a/AT 2017gfo [23], found in the host galaxy NGC 4993. The source of the signal was detected across the electromagnetic spectrum in the X-ray, ultraviolet, optical, infrared, and radio bands over hours, days, and weeks [24].

The electromagnetic transient SSS17a/AT 2017gfo was of great significance, as it provided the first compelling observational evidence for the nucleosynthesis of heavy nuclei through the r-process. As shown in Ref. [25], the observed emission was well reproduced by a three-component kilonova model: a lanthanide-poor blue component with an opacity of $\kappa = 0.5$ $cm^2\,g^{-1}$, ejecta mass $M_{\rm ej} \approx 0.020\,M_\odot$ and velocity $v_{\rm ej} = 0.27c$; an intermediate (purple) component with $\kappa = 3$ $cm^2\,g^{-1}$, $M_{\rm ej} \approx 0.047\,M_\odot$ and $v_{\rm ej} = 0.15c$; and a lanthanide-rich red component with $\kappa = 10$ $cm^2\,g^{-1}$, $M_{\rm ej} \approx 0.011\,M_\odot$ and $v_{\rm ej} = 0.14c$. Meanwhile, numerous efforts have been made to improve previous kilonova models, motivated by the prospect of obtaining high-resolution data from future observations with the James Webb Space Telescope (JWST). However, performing fully consistent general relativistic simulations that include a realistic treatment of nuclear matter with proper neutrino transport, followed by detailed nucleosynthesis calculations in the ejecta and subsequent radiative transfer simulations to reproduce the observed light curves, remains computationally demanding. Consequently, several practical semi-analytical formulations have been proposed to approximate key physical processes and reduce the computational cost while retaining the essential physics of the problem.

One of the cornerstones of analytical modeling was presented in Ref. [26], which introduced a simplified model not intended to replace full multi-group radiative transfer simulations, but rather to serve as a toy model for qualitatively understanding the evolution of the transient. More refined analytical approaches were proposed shortly thereafter. For instance, Ref. [27] developed a model capable of reproducing multi-band light curves by assuming a time-dependent effective opacity and a uniform power-law energy generation rate. Some models also account for the impact of ejecta morphology on the observed emission, introducing more realistic prescriptions for non-spherical geometries and their influence on the angular dependence of the kilonova light curves [28].

The present panorama of kilonova modeling extends beyond purely analytical formulations and is now better represented by semi-analytical frameworks, such as the one introduced in Ref. [29]. This framework provides a practical approach for generating broadband light curves that account for multiple ejecta components and non-spherical geometries, while incorporating time- and composition-dependent heating rates, thermalization efficiencies, and opacities.

Such advances in kilonova modeling have opened new possibilities for studying distinct features in the light curves that may contain additional information about the composition of the ejecta. It has been theorized that the late time kilonova emission, occurring weeks to months after the merger, can reveal the radioactive signatures of specific heavy nuclei, rather

than the power-law heating that dominates the early time emission [30]. Isotopes such as $^{2}23$Ra ($t_{1/2} = 11.43$ d) and $^{2}25$Ac ($t_{1/2} = 10.0$ d), which undergo $\alpha$-decay, and $^{2}54$Cf, which undergoes spontaneous fission, have been identified as potential dominant contributors to the late-time heating [31].

These nuclei release large amounts of energy per decay and the $\alpha$ particles thermalize more efficiently than typical $\beta$-decay products. Therefore, these nuclei are expected to produce detectable bumps or departures from a smooth power-law decline in the bolometric light curve. However, identifying such "fingerprints" requires sufficiently accurate heating-rate predictions from nucleosynthesis calculations and sensitive, multi-epoch, wide-band observations.

In March 2023, the electromagnetic transient AT 2023vfi, associated with the gamma-ray burst GRB 230307A, was identified as the second confirmed observation of a kilonova [32]. This signal has provided an early opportunity to apply these ideas observationally [33]. JWST follow-up analyses indicate that AT 2023vfi's early evolution resembles AT 2017gfo during the first week, and detailed spectral/photometric modeling has been used to place constraints on the presence and abundances of heavy nuclei in the ejecta. The work presented in Ref. [34] reports upper limits and tentative abundance estimates for isotopes such as $^{222}$Rn, $^{223}$Ra, $^{224}$Ra, and $^{225}$Ac. These results illustrate both the promise and the current limitations: while JWST data can probe late-time radioactive contributors, robust identification of specific isotopes requires careful propagation of nuclear-physics (decay-rate and fission-yield) and opacity uncertainties. In this context, one of the goals of this work is to assess whether a more accurate treatment of decay-rate uncertainties can improve the constraints on the abundance of heavy nuclei.

## 1.4 Objectives and thesis structure

As mentioned before, the main objective of this work is to evaluate the sensitivity of kilonova light curves to the choice of theoretical decay rates prescription. We constructed an ensemble of possible initial thermodynamic trajectories within the ejecta. For each initial condition, we ran six nuclear reaction network calculations with three different $\beta^{-}$-decay and two $\alpha$-decay prescriptions. Then, with the results for the abundances and the heating rate, we modeled the kilonova. We found consistent deviations from the power-law behavior across the ensemble of initial conditions. We also identified the nuclei responsible for such behavior along with their decay chains.

The thesis is divided into three main parts. The introductory part are Chapters 1 and 2. Chapter 2 develops the physics needed to understand neutron stars and binary neutron stars. Chapters 3, 4, and 5 present the basic theoretical framework for the three stages of the merger simulation: the general-relativistic hydrodynamics, the nuclear reaction network evolution, and the construction of the kilonova light curve from emission models. The last part, chapters 6, 7 and 9, will explain the details of the construction of the initial thermodynamic trajectories, the setup of the nuclear reaction network calculations with their results and the construction of the kilonova light curves.

# Chapter 2

# From Neutron stars to Kilonovas

Kilonovae is going to be our main topic of study through the whole thesis. Every chapter and section was build with the objective to develop the intuition on all the processes that takes place in the kilonova. However a necessary astrophysical context will enrich our understanding. This chapter therefore will first present the neutron star with its dense matter in the general relativity description. Then the text will move to the problem of finding two neutron star in a close enough orbit and present a little discussion on why the existing of such configurations are not trivial. After that, a description of the different phases of the merger will be presented. The different emission mechanisms will be discussed, gravitational waves, short gamma ray bursts and our favorite the kilonova.

## 2.1 The Neutron star

Neutron stars contain the densest matter known to exist in a stable state. Their central densities reach several times the nuclear saturation density $n_{\rm sat} \simeq 0.15$ fm$^{-3}$ [35]. In the outer layers of the star, nuclei coexist with a degenerate electron gas. With increasing density, electron captures make the nuclei progressively more neutron-rich until, at the neutron-drip density $n_{\rm drip} \simeq 2.5 \times 10^{-4}$ fm$^{-3}$ ($\rho \simeq 4 \times 10^{11}$ g cm$^{-3}$), neutrons are no longer bound to the nuclei and form a free neutron gas [36]. At about half the saturation density the difference between inside and outside of a nucleus vanish effectively becoming a continuum of nucleons which marks the transition from crust to core. At the same time the $\beta$ equilibrium favors neutrons instead of protons, this leads to a neutron dominated composition.

Observed neutron-star masses range from about 1.1 to slightly above $2\,M_\odot$, and their radii are inferred to lie between roughly 10 and 14 km [37,38]. However, a free degenerate neutron gas can support stars of at most $\simeq 0.7\,M_\odot$ [39,40]. To accurate explain observations the strongly repulsive short-range part of the nuclear interaction is required. The EOS, therefore, will have to include nuclear interaction terms which is quite a formidable task to calculate.

The microscopic theory of this matter, quantum chromodynamics (QCD), cannot at present provide the EOS in the relevant density range. At the baryon chemical potentials of neutron-star cores, QCD is non-perturbative, and lattice QCD is hampered by the fermion sign problem at finite baryon density. Controlled calculations exist only in two limits, chiral effective field theory describes nuclear matter up to about 1-2 $n_{\rm sat}$ [41], and perturbative

QCD becomes reliable only at densities of order $40\,n_{\rm sat}$, far above those realized in neutron stars [42]. In the intermediate regime, where neutron-star cores reside, the EOS is described by phenomenological models. As a consequence, a description from first principles of neutron star cores remains unknown. Matter at several times $n_{\rm sat}$ may consist of nucleons, may contain hyperons, or may undergo a transition to deconfined quark matter, either through a smooth crossover or through a first order hadron-quark phase transition [43].

Lets assume for instance that we know the EOS, then can estimate the radius of the star for a certain range of masses including GR effects through the Tolman-Oppenheimer-Volkoff equation. the equation assumes a statics spherical symmetric perfect, fluid which is completely fine but real stars have angular momentum so rotational corrections will be relevant.

$$\frac{dP}{dr} = -\frac{G}{r^2}\left(\rho + \frac{P}{c^2}\right)\left(m + 4\pi r^3 \frac{P}{c^2}\right)\left(1 - \frac{2Gm}{c^2 r}\right)^{-1} \tag{2.1}$$

Here $\rho(r)$ and $P(r)$ are density and pressure at a radial distance $r$. The quantity $m(r)$ is the mass within a sphere of radius $r$. At this point we have not introduce any EOS yet. The relationship between $P$ and $\rho$ could depend on temperature and composition. Many current models for EOS actually presents their results for many temperatures and $Y_e$. Once we stablish the EOS one can find a radius-mass curve. By comparing with observations one can in principle infer constrains on the EOS. Such studies have been done, for example existence of pulsars with masses of about $2\,M_\odot$ [44] and the radii inferred by NICER [38] constrain the pressure of dense matter up to densities of roughly $2-3\,n_{\rm sat}$ [45]. These measurements, however, constrain the relation between pressure and energy density rather than giving a full description of the matter. For example, hybrid stars (Neutron stars with quark cores) can have mass-radius relations nearly indistinguishable from those of purely hadronic stars [46].

## 2.2 Formation of binary neutron stars

For two neutron stars to merge, three conditions must be met. Both stars must form, which requires two core-collapse supernovae so the binary must survive both explosions. Then the two remnants must end up close enough for gravitational-wave (GW) emission to bring them into contact in less than a Hubble time. For typical neutron-star masses, the initial separation must be smaller than about $5\,R_\odot$. This is two to three orders of magnitude smaller than the radii that the progenitor stars reach as supergiants. The central problem of binary-neutron-star formation is therefore how to bring the compact remnants of two massive stars to separations far smaller than the stars themselves once were, without unbinding the system on the way.

In the standard isolated channel, a double neutron star (DNS) descends from a binary of two OB stars that are massive enough to end their lives in a core-collapse SN, see Fig. 2.1. If the initial binary are separated enough the descending DNS will not be close enough to merge. If, on the contrary, the initial star are close, they will interact via stable or unstable mass transfer. Once one of the stars reach the end of its live the star will expands transferring part of its mass to their companion. If after the supernova the two stars remain bounded they will become a high-mass X-ray binary (HMXB). When the second star expands it will bolt the first star in a phase called common envelope [47]. In that stage the dynamical friction

of the motion of the first star inside the second one will cause the necessary loss of angular momentum to bring closer the cores. This phase is not stable so eventually the envelope will fall to the stars ofr get ejected leaving behind the first NS and the He core of the second one that eventually will die in what is called a ultra-strippend SN [48]. finally the system will results in a DNS close enough to merge. The sequence of events is reviewed in [49].

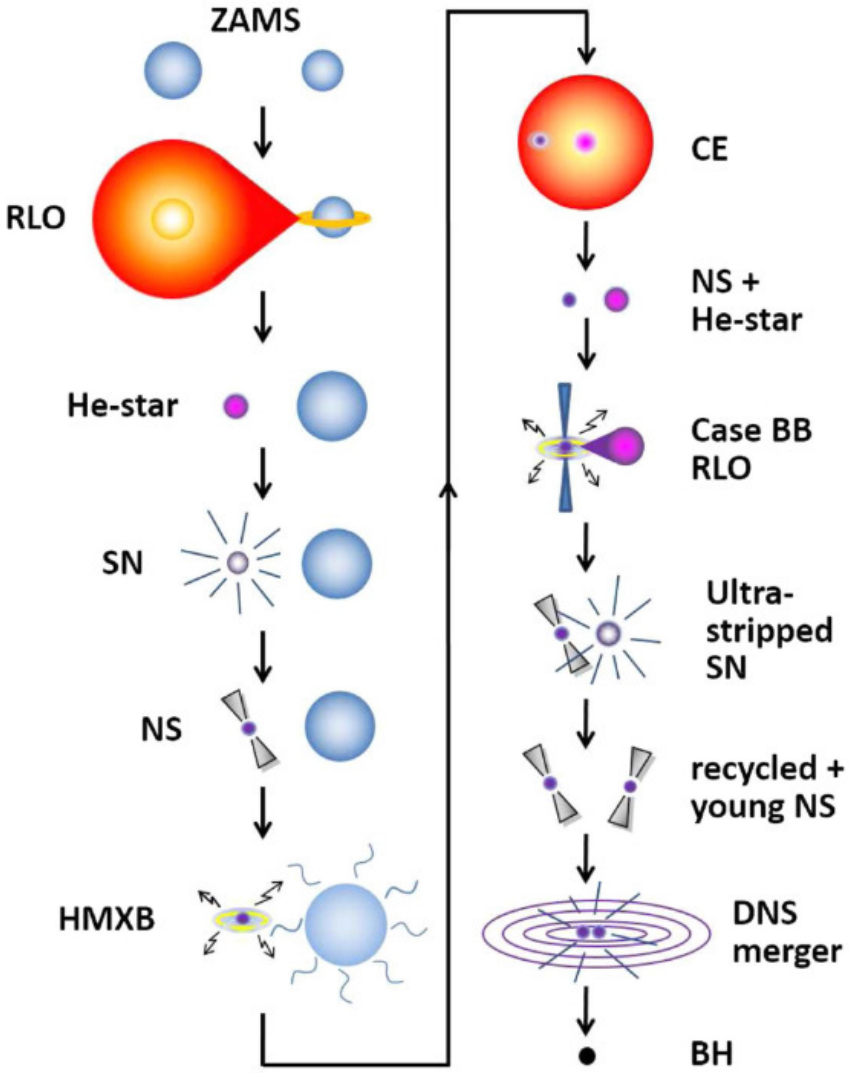



Figure 2.1: Double neutron star (DNS) formation from two OB stars as presented in [49]. ZAMS refers to zero-age main sequence, RLO roche lobe overflow, SN supernova, HMXB high mass X ray binary and CE common envelope

## 2.3 Gravitational-wave-driven inspiral and the maximum separation

Once both neutron stars have formed, the only process that changes the orbit appreciably is the emission of gravitational radiation. Here we will show an estimate of the time to the merger at leading (quadrupole) order and the maximum initial separation that allows a merger within the age of the Universe. This calculation is based on the work presented in [50].

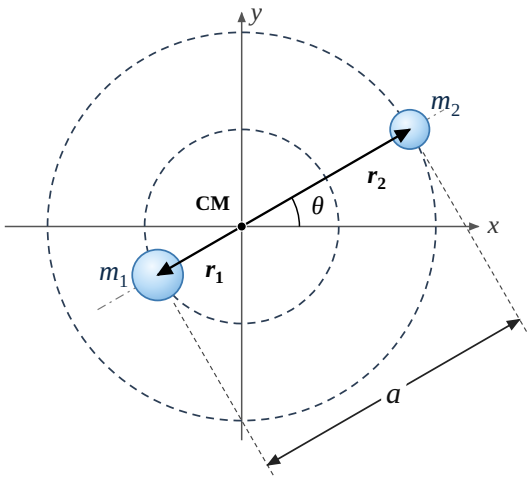



Figure 2.2: Two neutron stars with masses $m_1$ and $m_2$ orbiting at a distance $a$ seen from the center of mass frame.

Consider two point masses $m_1$ and $m_2$ on a circular orbit of separation $a$ and angular frequency $\omega = \sqrt{GM/a^3}$, with total mass $M = m_1 + m_2$ and reduced mass $\mu = m_1 m_2/M$. Their vector positions will be $\mathbf{r}_1$ and $\mathbf{r}_2$ respectively, see Fig 2.2. Define the quadrupole mass tensor as

$$Q_{ij} = \sum_l m_l r_{li} r_{lj}, \tag{2.2}$$

summing over the masses $m_l$ with $r_{li} = \mathbf{r}_l \cdot \hat{\imath}$. From this configuration the position are

$$\mathbf{r}_1 = -\frac{\mu a}{m_1}\hat{n}\,; \quad \mathbf{r}_2 = \frac{\mu a}{m_2}\hat{n} \quad \text{with} \quad \hat{n} = (cos(\theta), sin(\theta), 0). \tag{2.3}$$

So the quadrupole mass tensor components will be:

$$Q_{xx} = \frac{\mu^2 a^2}{m_1} cos^2(\theta) + \frac{\mu^2 a^2}{m_2} cos^2(\theta) = \mu a^2 cos^2(\theta) = \frac{\mu a^2}{2}(1 + cos(2\theta)) \tag{2.4}$$

$$Q_{yy} = \mu a^2 sin^2(\theta) = -\frac{\mu a^2}{2}(1 + cos(2\theta)) \tag{2.5}$$

$$Q_{xy} = \mu a^2 sin(\theta) cos(\theta) = \frac{\mu a^2}{2}(sin(2\theta)) \tag{2.6}$$

$$\tag{2.7}$$

For our fortune the power radiated in form of GW ($L_{GW}$) can be estimated using equation (6) from [50]

$$L_{\rm GW} = \frac{G}{5c^5}\left(\dddot{Q}_{ij}\dddot{Q}_{ij} - \frac{1}{3}\dddot{Q}_{ii}\dddot{Q}_{jj}\right) \tag{2.8}$$

$$L_{\rm GW} = \frac{G}{5c^5}\left(\dddot{Q}_{xx}{}^2 + \dddot{Q}_{yy}{}^2 + 2\dddot{Q}_{xy}{}^2 - \frac{1}{3}(\dddot{Q}_{xx} + \dddot{Q}_{yy})^2\right) \tag{2.9}$$

since $Q_{xx} = -Q_{yy}$ the last term vanishes and with $\dddot{Q}_{xx}^2 + \dddot{Q}_{yy}^2 + 2\dddot{Q}_{xy}^2 = 32\,\mu^2 a^4\omega^6$ (see appendix A), the GW luminosity gives

$$L_{\rm GW} = \frac{32}{5}\frac{G}{c^5}\mu^2 a^4\omega^6 = \frac{32}{5}\frac{G^4}{c^5}\frac{(m_1 m_2)^2 M}{a^5}. \tag{2.10}$$

The energy radiated is drawn from the orbital energy $E = -Gm_1m_2/(2a)$. Setting $\mathrm{d}E/\mathrm{d}t = -L_{\rm GW}$ gives

$$\frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{\beta}{a^3}, \qquad \beta \equiv \frac{64}{5}\frac{G^3 m_1 m_2 M}{c^5}. \tag{2.11}$$

Equation (2.11) integrates to $a^4(t) = a_0^4 - 4\beta t$, so a binary with initial separation $a_0$ merge ($a \to 0$) after

$$t_m(a_0) = \frac{a_0^4}{4\beta} = \frac{5}{256}\frac{c^5 a_0^4}{G^3 m_1 m_2 M}. \tag{2.12}$$

The steep $a_0^4$ dependence is the key point. Halving the separation shortens the inspiral by a factor of sixteen.

Now lets estimate the maximum separation for BNS to merger within the age of the universe. Notice that since the stars require a preliminary formation and evolution we will therefore over estimate the maximum distances between stars. Lets require $t_m \le t_{\rm H}$, with $t_{\rm H} \simeq 13.8$ Gyr the age of the Universe

$$a_{\rm max} = (4\beta\, t_{\rm H})^{1/4} = \left(\frac{256}{5}\frac{G^3 m_1 m_2 M}{c^5}\, t_{\rm H}\right)^{1/4}. \tag{2.13}$$

For $m_1 = m_2 = 1.35\,M_\odot$, $m_1 m_2 M = 4.92\,M_\odot^3$ and $\beta = 6.08 \times 10^{27}$ cm$^4$ s$^{-1}$. Hence

$$a_{\rm max} \simeq 3.2 \times 10^6 \text{ km} \simeq 4.6\,R_\odot, \tag{2.14}$$

about 0.02 AU. As reference red giants have typical radius of hundreds of $R_\odot$ so having neutron stars closer than $4.6\,R_\odot$ is quite a challenge.

## 2.4 Mass ejection: tidal, shock-driven, and disk-wind ejecta

## 2.5 Weak interactions and the electron fraction of the ejecta

## 2.6 Electromagnetic counterparts

# Chapter 3

# Numerical Relativistic Hydrodynamics

**3.1 3+1 decomposition and evolution formulation**

**3.2 General-relativistic hydrodynamics in conservative form**

**3.3 The Einstein Toolkit: initial data, mesh refinement, atmosphere**

**3.4 Gravitational-wave extraction**

**3.5 Ejecta diagnostics and unbound-matter criteria**

# Chapter 4

# r-Process Nucleosynthesis

**4.1 Nuclear statistical equilibrium and freeze-out**

**4.2 The r-process: neutron-to-seed ratio, fission cycling**

**4.3 Reaction-network equations and numerical solution**

**4.4 Nuclear inputs: masses, decay rates, neutron capture, fission**

**4.5 Radioactive heating and thermalization**

# Chapter 5

# Kilonova Light Curves

## 5.1 Li-Paczyński model

## 5.2 Radiative transfer and the diffusion regime

## 5.3 Opacities of r-process matter

## 5.4 Light-curve modeling

# Chapter 6

# Methodology on the impact of theoretical decay rates across thermodynamics trajectories

## 6.1 Binary neutron-star simulation

## 6.2 From simulation to trajectories: ejecta extraction and expansion

## 6.3 Initial termodinamic trajectories treatment

## 6.4 Nuclear reaction network setup and theoretical reaction reates

## 6.5 Modified Li-Paczyński model and radiative transfer simulations

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

# Chapter 7

# Results

## 7.1 Merger dynamics

## 7.2 Dynamical-ejecta properties

## 7.3 Nucleosynthesis and heating rates

## 7.4 Kilonova light curves

## 7.5 Relevant nuclei and uncertainties

## 7.6 Detectability with current and future telescopes

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

# Chapter 8

# Discussion

## 8.1 Comparison with previous studies

## 8.2 Limitations and model assumptions

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

# Chapter 9

# Conclusions and Outlook

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

# Appendix A

# Quadrupole and $L_{GW}$

In this appendix we will go through the details of finding the mass quadrupole components, explain the approximations assumed and finding the $L_{GW}$. Lets start with the definitions, as we saw from Fig. 2.2 the position are:

$$\mathbf{r}_1 = -\frac{\mu a}{m_1}(cos(\theta), sin(\theta), 0), \tag{A.1}$$

$$\mathbf{r}_2 = \frac{\mu a}{m_2}(cos(\theta), sin(\theta), 0). \tag{A.2}$$

$$\tag{A.3}$$

We used $||\mathbf{r}_1|| = m_2 a/(m_1 + m_2) = \mu a/m_1$ with $\mu = m_1 m_2/(m_1 + m_2)$. The first quadrupole component will be:

$$Q_{xx} = m_1 r_{x1} r_{x1} + m_2 r_{x2} r_{x2} = m_1 \frac{\mu^2 a^2}{m_1^2} cos^2(\theta) + m_2 \frac{\mu^2 a^2}{m_2^2} cos^2(\theta), \tag{A.4}$$

$$Q_{xx} = \mu^2 a^2 cos^2(\theta)(\frac{1}{m_1} + \frac{1}{m_2}), \tag{A.5}$$

$$Q_{xx} = \mu a^2 cos^2(\theta), \tag{A.6}$$

$$Q_{xx} = \frac{\mu a^2}{2}(cos(2\theta) + 1). \tag{A.7}$$

$$\tag{A.8}$$

Similarly for $Q_{yy}$:

$$Q_{yy} = m_1 r_{y1} r_{y1} + m_2 r_{y2} r_{y2} = m_1 \frac{\mu^2 a^2}{m_1^2} sin^2(\theta) + m_2 \frac{\mu^2 a^2}{m_2^2} sin^2(\theta), \tag{A.9}$$

$$Q_{yy} = \mu^2 a^2 sin^2(\theta)(\frac{1}{m_1} + \frac{1}{m_2}), \tag{A.10}$$

$$Q_{yy} = \mu a^2 sin^2(\theta), \tag{A.11}$$

$$Q_{yy} = \frac{\mu a^2}{2}(-cos(2\theta) + 1). \tag{A.12}$$

$$\tag{A.13}$$

Finally for $Q_{xy}$

$$Q_{xy} = m_1 r_{x1} r_{y1} + m_2 r_{x2} r_{y2} = m_1 \frac{\mu^2 a^2}{m_1^2} sin(\theta) cos(\theta) + m_2 \frac{\mu^2 a^2}{m_2^2} sin(\theta) cos(\theta), \tag{A.14}$$

$$Q_{xy} = \mu^2 a^2 sin(\theta) cos(\theta) (\frac{1}{m_1} + \frac{1}{m_2}), \tag{A.15}$$

$$Q_{xy} = \mu a^2 sin(\theta) cos(\theta), \tag{A.16}$$

$$Q_{xy} = \frac{\mu a^2}{2} sin(2\theta). \tag{A.17}$$

$$\tag{A.18}$$

Now we need the third derivate with respect to time. We will assume two things, $\ddot{\theta} << 1$ and $\dot{a} << 1$. Effectively we are saying that the change in angular velocity and the change in the radius is slow which is referred as an adiabatic process. To see how good the approximation is we can compare the orbit time scale with the inspiral timescale. Which in fact is the situation, therefore is a good approximation until the merger itself. Therefore it is easy to see that:

$$\dddot{Q}_{xx} = 4\mu a^2 \dot{\theta}^3 sin(2\theta) \tag{A.19}$$

$$\dddot{Q}_{yy} = -4\mu a^2 \dot{\theta}^3 sin(2\theta) \tag{A.20}$$

$$\dddot{Q}_{xy} = -4\mu a^2 \dot{\theta}^3 cos(2\theta) \tag{A.21}$$

$$\tag{A.22}$$

Now we meed to calculate the following term remembering that we are using einstein notation for the repeated indices and that $Q_{xy} = Q_{yx}$:

$$\dddot{Q}_{ij} \dddot{Q}_{ij} - \dddot{Q}_{ii} \dddot{Q}_{jj} = \dddot{Q}_{xx} \dddot{Q}_{xx} + \dddot{Q}_{xy} \dddot{Q}_{xy} + \dddot{Q}_{yx} \dddot{Q}_{yx} + \dddot{Q}_{yy} \dddot{Q}_{yy} \tag{A.23}$$

$$- \dddot{Q}_{xx} \dddot{Q}_{xx} - \dddot{Q}_{yy} \dddot{Q}_{xx} - \dddot{Q}_{xx} \dddot{Q}_{yy} - \dddot{Q}_{yy} \dddot{Q}_{yy} \tag{A.24}$$

$$= \dddot{Q}_{xx}^2 + 2\dddot{Q}_{xy}^2 + \dddot{Q}_{yy}^2 - \dddot{Q}_{xx}^2 - 2\dddot{Q}_{yy} \dddot{Q}_{xx} - \dddot{Q}_{yy}^2 \tag{A.25}$$

$$= \dddot{Q}_{xx}^2 + 2\dddot{Q}_{xy}^2 + \dddot{Q}_{yy}^2 - (\dddot{Q}_{xx} + \dddot{Q}_{yy})^2 \tag{A.26}$$

$$= 16\mu^2 a^4 \dot{\theta}^6 (sin^2(2\theta) + 2cos^2(2\theta) + sin^2(2\theta)) \tag{A.27}$$

$$= 32\mu^2 a^4 \dot{\theta}^6 = \frac{32 G^3 m_1^2 m_2^2 M}{a^5} \tag{A.28}$$

Finally:

$$L_{GW} = \frac{G}{5c^5} (\dddot{Q}_{ij} \dddot{Q}_{ij} - \dddot{Q}_{ii} \dddot{Q}_{jj}) = \frac{32 G^4 m_1^2 m_2^2 M}{c^5 a^5} \tag{A.29}$$

# Appendix B

# Convergence tests

# Appendix C

# Network and equation-of-state configuration details

# References


[1] J. J. Cowan, C. Sneden, J. E. Lawler, A. Aprahamian, M. Wiescher, K. Langanke, G. Martínez-Pinedo, and F.-K. Thielemann. "Origin of the heaviest elements: The rapid neutron-capture process." *Reviews of Modern Physics*, **93**(1), Feb. 1, 2021, p. 015002. ISSN: 0034-6861, 1539-0756. DOI: 10.1103/RevModPhys.93.015002. URL: https://link.aps.org/doi/10.1103/RevModPhys.93.015002.

[2] S. Rosswog and O. Korobkin. "Heavy Elements and Electromagnetic Transients from Neutron Star Mergers." *Annalen der Physik*, **536**(2), Feb. 2024, p. 2200306. ISSN: 0003-3804, 1521-3889. DOI: 10.1002/andp.202200306. URL: https://onlinelibrary.wiley.com/doi/10.1002/andp.202200306 (visited on 11/06/2025).

[3] B. D. Metzger, G. Martínez-Pinedo, S. Darbha, E. Quataert, A. Arcones, D. Kasen, R. Thomas, P. Nugent, I. V. Panov, and N. T. Zinner. "Electromagnetic counterparts of compact object mergers powered by the radioactive decay of r-process nuclei: Transients from compact object mergers." *Monthly Notices of the Royal Astronomical Society*, **406**(4), Aug. 21, 2010, pp. 2650–2662. ISSN: 00358711. DOI: 10.1111/j.1365-2966.2010.16864.x. URL: https://academic.oup.com/mnras/article-lookup/doi/10.1111/j.1365-2966.2010.16864.x (visited on 10/15/2025).

[4] J. Barnes, D. Kasen, M.-R. Wu, and G. Martínez-Pinedo. "Radioactivity and thermalization in the ejecta of compact object mergers and their impact on kilonova light curves." *The Astrophysical Journal*, **829**(2), Oct. 2016, p. 110. ISSN: 0004-637X, 1538-4357. DOI: 10.3847/0004-637X/829/2/110. URL: https://iopscience.iop.org/article/10.3847/0004-637X/829/2/110.

[5] Y.-H. Yang et al. "A lanthanide-rich kilonova in the aftermath of a long gamma-ray burst." en. *Nature*, **626**(8000), Feb. 2024, pp. 742–745. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-023-06979-5. URL: https://www.nature.com/articles/s41586-023-06979-5.

[6] R. H. Cyburt, B. D. Fields, K. A. Olive, and T.-H. Yeh. "Big bang nucleosynthesis: Present status." en. *Reviews of Modern Physics*, **88**(1), Feb. 2016, p. 015004. ISSN: 0034-6861, 1539-0756. DOI: 10.1103/RevModPhys.88.015004. URL: https://link.aps.org/doi/10.1103/RevModPhys.88.015004.

[7] F.-K. Thielemann, J. Isern, A. Perego, and P. Von Ballmoos. "Nucleosynthesis in Supernovae." en. *Space Science Reviews*, **214**(3), Apr. 2018, p. 62. ISSN: 0038-6308, 1572-9672. DOI: 10.1007/s11214-018-0494-5. URL: http://link.springer.com/10.1007/s11214-018-0494-5.

[8] E. M. Burbidge, G. R. Burbidge, W. A. Fowler, and F. Hoyle. "Synthesis of the elements in stars." en. *Reviews of Modern Physics*, **29**(4), Oct. 1957, pp. 547–650. ISSN: 0034-6861. DOI: 10.1103/RevModPhys.29.547. URL: https://link.aps.org/doi/10.1103/RevModPhys.29.547.

[9] F. Käppeler, R. Gallino, S. Bisterzo, and W. Aoki. "The s process: Nuclear physics, stellar models, and observations." en. *Reviews of Modern Physics*, **83**(1), Apr. 2011, pp. 157–193. ISSN: 0034-6861, 1539-0756. DOI: 10.1103/RevModPhys.83.157. URL: https://link.aps.org/doi/10.1103/RevModPhys.83.157.

[10] R. Hoffman, S. Woosley, and Y.-Z. Qian. "Model independent r-process nucleosynthesis — Constraints on the key parameters." en. *Nuclear Physics A*, **621**(1-2), Aug. 1997, pp. 397–400. ISSN: 03759474. DOI: 10.1016/S0375-9474(97)00278-9. URL: https://linkinghub.elsevier.com/retrieve/pii/S0375947497002789.

[11] S. E. Woosley, J. R. Wilson, G. J. Mathews, R. D. Hoffman, and B. S. Meyer. "The r-process and neutrino-heated supernova ejecta." en. *The Astrophysical Journal*, **433**, Sept. 1994, p. 229. ISSN: 0004-637X, 1538-4357. DOI: 10.1086/174638. URL: http://adsabs.harvard.edu/doi/10.1086/174638.

[12] A. Arcones, H.-T. Janka, and L. Scheck. "Nucleosynthesis-relevant conditions in neutrino-driven supernova outflows: I. Spherically symmetric hydrodynamic simulations." *Astronomy & Astrophysics*, **467**(3), June 2007, pp. 1227–1248. ISSN: 0004-6361, 1432-0746. DOI: 10.1051/0004-6361:20066983. URL: http://www.aanda.org/10.1051/0004-6361:20066983.

[13] C. Sneden, J. J. Cowan, and R. Gallino. "Neutron-Capture Elements in the Early Galaxy." en. *Annual Review of Astronomy and Astrophysics*, **46**(1), Sept. 2008, pp. 241–288. ISSN: 0066-4146, 1545-4282. DOI: 10.1146/annurev.astro.46.060407.145207. URL: https://www.annualreviews.org/doi/10.1146/annurev.astro.46.060407.145207.

[14] A. P. Ji, A. Frebel, J. D. Simon, and A. Chiti. "COMPLETE ELEMENT ABUNDANCES OF NINE STARS IN THE r-PROCESS GALAXY RETICULUM II*." *The Astrophysical Journal*, **830**(2), Oct. 2016, p. 93. ISSN: 0004-637X, 1538-4357. DOI: 10.3847/0004-637X/830/2/93. URL: https://iopscience.iop.org/article/10.3847/0004-637X/830/2/93.

[15] A. Wallner et al. "Abundance of live 244Pu in deep-sea reservoirs on Earth points to rarity of actinide nucleosynthesis." en. *Nature Communications*, **6**(1), Jan. 2015, p. 5956. ISSN: 2041-1723. DOI: 10.1038/ncomms6956. URL: https://www.nature.com/articles/ncomms6956.

[16] J. M. Lattimer and D. N. Schramm. "The tidal disruption of neutron stars by black holes in close binaries." en. *The Astrophysical Journal*, **210**, Dec. 1976, p. 549. ISSN: 0004-637X, 1538-4357. DOI: 10.1086/154860. URL: http://adsabs.harvard.edu/doi/10.1086/154860.

[17] L.-X. Li and B. Paczyński. "Transient Events from Neutron Star Mergers." *The Astrophysical Journal*, **507**(1), Nov. 1, 1998, pp. L59–L62. ISSN: 0004637X. DOI: 10.1086/311680. URL: https://iopscience.iop.org/article/10.1086/311680 (visited on 10/14/2025).

[18] D. Kasen, N. R. Badnell, and J. Barnes. "OPACITIES AND SPECTRA OF THE $r$ -PROCESS EJECTA FROM NEUTRON STAR MERGERS." *The Astrophysical Journal*, **774**(1), Aug. 12, 2013, p. 25. ISSN: 0004-637X, 1538-4357. DOI: 10.1088/0004-637X/774/1/25. URL: https://iopscience.iop.org/article/10.1088/0004-637X/774/1/25.

[19] M. Tanaka and K. Hotokezaka. "RADIATIVE TRANSFER SIMULATIONS OF NEUTRON STAR MERGER EJECTA." *The Astrophysical Journal*, **775**(2), Sept. 13, 2013, p. 113. ISSN: 0004-637X, 1538-4357. DOI: 10.1088/0004-637X/775/2/113. URL: https://iopscience.iop.org/article/10.1088/0004-637X/775/2/113.

[20] J. Barnes and D. Kasen. "EFFECT OF A HIGH OPACITY ON THE LIGHT CURVES OF RADIOACTIVELY POWERED TRANSIENTS FROM COMPACT OBJECT MERGERS." *The Astrophysical Journal*, **775**(1), Aug. 29, 2013, p. 18. ISSN: 0004-637X, 1538-4357. DOI: 10.1088/0004-637X/775/1/18. URL: https://iopscience.iop.org/article/10.1088/0004-637X/775/1/18.

[21] J. Lippuner and L. F. Roberts. "r-PROCESS LANTHANIDE PRODUCTION AND HEATING RATES IN KILONOVAE." *The Astrophysical Journal*, **815**(2), Dec. 11, 2015, p. 82. ISSN: 1538-4357. DOI: 10.1088/0004-637X/815/2/82. URL: https://iopscience.iop.org/article/10.1088/0004-637X/815/2/82 (visited on 10/27/2025).

[22] B. P. Abbott et al. "GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral." *Physical Review Letters*, **119**(16), Oct. 16, 2017, p. 161101. ISSN: 0031-9007, 1079-7114. DOI: 10.1103/PhysRevLett.119.161101. URL: https://link.aps.org/doi/10.1103/PhysRevLett.119.161101.

[23] D. A. Coulter et al. "Swope Supernova Survey 2017a (SSS17a), the optical counterpart to a gravitational wave source." *Science*, **358**(6370), Dec. 22, 2017, pp. 1556–1558. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.aap9811. URL: https://www.science.org/doi/10.1126/science.aap9811.

[24] M. R. Drout et al. "Light curves of the neutron star merger GW170817/SSS17a: Implications for r-process nucleosynthesis." *Science*, **358**(6370), Dec. 22, 2017, pp. 1570–1574. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.aaq0049. URL: https://www.science.org/doi/10.1126/science.aaq0049.

[25] V. A. Villar et al. "The Combined Ultraviolet, Optical, and Near-infrared Light Curves of the Kilonova Associated with the Binary Neutron Star Merger GW170817: Unified Data Set, Analytic Models, and Physical Implications." *The Astrophysical Journal Letters*, **851**(1), Dec. 10, 2017, p. L21. ISSN: 2041-8205, 2041-8213. DOI: 10.3847/2041-8213/aa9c84. URL: https://iopscience.iop.org/article/10.3847/2041-8213/aa9c84.

[26] B. D. Metzger. "Kilonovae." *Living Reviews in Relativity*, **20**(1), Dec. 2017, p. 3. ISSN: 2367-3613, 1433-8351. DOI: 10.1007/s41114-017-0006-z. URL: http://link.springer.com/10.1007/s41114-017-0006-z.

[27] E. Waxman, E. O. Ofek, D. Kushnir, and A. Gal-Yam. "Constraints on the ejecta of the GW170817 neutron star merger from its electromagnetic emission." *Monthly Notices of the Royal Astronomical Society*, **481**(3), Dec. 11, 2018, pp. 3423–3441. ISSN: 0035-8711, 1365-2966. DOI: 10.1093/mnras/sty2441. URL: https://academic.oup.com/mnras/article/481/3/3423/5091828.

[28] R. T. Wollaeger, O. Korobkin, C. J. Fontes, S. K. Rosswog, W. P. Even, C. L. Fryer, J. Sollerman, A. L. Hungerford, D. R. van Rossum, and A. B. Wollaber. "Impact of ejecta morphology and composition on the electromagnetic signatures of neutron star mergers." *Monthly Notices of the Royal Astronomical Society*, **478**(3), Aug. 11, 2018, pp. 3298–3334. ISSN: 0035-8711, 1365-2966. DOI: 10.1093/mnras/sty1018. URL: https://academic.oup.com/mnras/article/478/3/3298/4980938.

[29] G. Ricigliano, A. Perego, S. Borhanian, E. Loffredo, K. Kawaguchi, S. Bernuzzi, and L. C. Lippold. "xkn: a semi-analytic framework for the modelling of kilonovae." *Monthly Notices of the Royal Astronomical Society*, **529**(1), Feb. 27, 2024, pp. 647–663. ISSN: 0035-8711, 1365-2966. DOI: 10.1093/mnras/stae572. URL: https://academic.oup.com/mnras/article/529/1/647/7613575.

[30] M.-R. Wu, J. Barnes, G. Martínez-Pinedo, and B. D. Metzger. "Fingerprints of Heavy-Element Nucleosynthesis in the Late-Time Lightcurves of Kilonovae." *Physical Review Letters*, **122**(6), Feb. 12, 2019, p. 062701. ISSN: 0031-9007, 1079-7114. DOI: 10.1103/PhysRevLett.122.062701. URL: https://link.aps.org/doi/10.1103/PhysRevLett.122.062701.

[31] M.-H. Chen and E.-W. Liang. "Radioactive decay of specific heavy elements as an energy source for late-time kilonovae and potential *James Webb Space Telescope* observations." *Monthly Notices of the Royal Astronomical Society*, **527**(3), Nov. 27, 2023, pp. 5540–5546. ISSN: 0035-8711, 1365-2966. DOI: 10.1093/mnras/stad3523. URL: https://academic.oup.com/mnras/article/527/3/5540/7424991.

[32] A. J. Levan et al. "Heavy-element production in a compact object merger observed by JWST." *Nature*, **626**(8000), Feb. 22, 2024, pp. 737–741. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-023-06759-1. URL: https://www.nature.com/articles/s41586-023-06759-1.

[33] J. H. Gillanders and S. J. Smartt. "Analysis of the *JWST* spectra of the kilonova AT 2023vfi accompanying GRB 230307A." *Monthly Notices of the Royal Astronomical Society*, **538**(3), Mar. 17, 2025, pp. 1663–1689. ISSN: 0035-8711, 1365-2966. DOI: 10.1093/mnras/staf287. URL: https://academic.oup.com/mnras/article/538/3/1663/8019796.

[34] Z. Liu, J.-a. Jiang, and W. Zhao. "A Constraint on Superheavy Elements of the GRB–Kilonova AT 2023vfi." *The Astrophysical Journal Letters*, **983**(2), Apr. 20, 2025, p. L34. ISSN: 2041-8205, 2041-8213. DOI: 10.3847/2041-8213/adc260. URL: https://iopscience.iop.org/article/10.3847/2041-8213/adc260.

[35] C. J. Horowitz, J. Piekarewicz, and B. Reed. "Insights into nuclear saturation density from parity-violating electron scattering." en. *Physical Review C*, **102**(4), Oct. 2020, p. 044321. ISSN: 2469-9985, 2469-9993. DOI: 10.1103/PhysRevC.102.044321. URL: https://link.aps.org/doi/10.1103/PhysRevC.102.044321.

[36] G. Baym, H. A. Bethe, and C. J. Pethick. "Neutron star matter." en. *Nuclear Physics A*, **175**(2), Nov. 1971, pp. 225–271. ISSN: 03759474. DOI: 10.1016/0375-9474(71)90281-8. URL: https://linkinghub.elsevier.com/retrieve/pii/0375947471902818.

[37] F. Özel and P. Freire. "Masses, Radii, and the Equation of State of Neutron Stars." en. *Annual Review of Astronomy and Astrophysics*, **54**(1), Sept. 2016, pp. 401–440. ISSN: 0066-4146, 1545-4282. DOI: 10.1146/annurev-astro-081915-023322. URL: https://www.annualreviews.org/content/journals/10.1146/annurev-astro-081915-023322.

[38] T. E. Riley et al. "A NICER View of the Massive Pulsar PSR J0740+6620 Informed by Radio Timing and XMM-Newton Spectroscopy." *The Astrophysical Journal Letters*, **918**(2), Sept. 2021, p. L27. ISSN: 2041-8205, 2041-8213. DOI: 10.3847/2041-8213/ac0a81. URL: https://iopscience.iop.org/article/10.3847/2041-8213/ac0a81.

[39] J. R. Oppenheimer and G. M. Volkoff. "On Massive Neutron Cores." en. *Physical Review*, **55**(4), Feb. 1939, pp. 374–381. ISSN: 0031-899X. DOI: 10.1103/PhysRev.55.374. URL: https://link.aps.org/doi/10.1103/PhysRev.55.374.

[40] R. C. Tolman. "Static Solutions of Einstein's Field Equations for Spheres of Fluid." en. *Physical Review*, **55**(4), Feb. 1939, pp. 364–373. ISSN: 0031-899X. DOI: 10.1103/PhysRev.55.364. URL: https://link.aps.org/doi/10.1103/PhysRev.55.364 (visited on 10/29/2026).

[41] K. Hebeler, J. M. Lattimer, C. J. Pethick, and A. Schwenk. "Equation of state and neutron star properties constrained by nuclear physics." *The Astrophysical Journal*, **773**(1), July 2013, p. 11. ISSN: 0004-637X, 1538-4357. DOI: 10.1088/0004-637X/773/1/11. URL: https://iopscience.iop.org/article/10.1088/0004-637X/773/1/11.

[42] A. Kurkela, E. S. Fraga, J. Schaffner-Bielich, and A. Vuorinen. "Constraining neutron star matter with quantum chromodynamics." *The Astrophysical Journal*, **789**(2), June 2014, p. 127. ISSN: 0004-637X, 1538-4357. DOI: 10.1088/0004-637X/789/2/127. URL: https://iopscience.iop.org/article/10.1088/0004-637X/789/2/127.

[43] G. Baym, T. Hatsuda, T. Kojo, P. D. Powell, Y. Song, and T. Takatsuka. "From hadrons to quarks in neutron stars: a review." *Reports on Progress in Physics*, **81**(5), May 2018, p. 056902. ISSN: 0034-4885, 1361-6633. DOI: 10.1088/1361-6633/aaae14. URL: https://iopscience.iop.org/article/10.1088/1361-6633/aaae14.

[44] E. Fonseca et al. "Refined Mass and Geometric Measurements of the High-mass PSR J0740+6620." *The Astrophysical Journal Letters*, **915**(1), July 2021, p. L12. ISSN: 2041-8205, 2041-8213. DOI: 10.3847/2041-8213/ac03b8. URL: https://iopscience.iop.org/article/10.3847/2041-8213/ac03b8.

[45] G. Raaijmakers, S. K. Greif, K. Hebeler, T. Hinderer, S. Nissanke, A. Schwenk, T. E. Riley, A. L. Watts, J. M. Lattimer, and W. C. G. Ho. "Constraints on the Dense Matter Equation of State and Neutron Star Properties from NICER's Mass–Radius Estimate of PSR J0740+6620 and Multimessenger Observations." *The Astrophysical Journal Letters*, **918**(2), Sept. 2021, p. L29. ISSN: 2041-8205, 2041-8213. DOI: 10.3847/2041-8213/ac089a. URL: https://iopscience.iop.org/article/10.3847/2041-8213/ac089a.

[46] M. Alford, M. Braby, M. Paris, and S. Reddy. "Hybrid Stars that Masquerade as Neutron Stars." en. *The Astrophysical Journal*, **629**(2), Aug. 2005, pp. 969–978. ISSN: 0004-637X, 1538-4357. DOI: 10.1086/430902. URL: https://iopscience.iop.org/article/10.1086/430902.

[47] B. Paczynski. "Common Envelope Binaries." en. *Symposium - International Astronomical Union*, **73**, 1976, pp. 75–80. ISSN: 0074-1809. DOI: 10.1017/S0074180900011864. URL: https://www.cambridge.org/core/product/identifier/S0074180900011864/type/journal_article (visited on 10/27/2026).

[48] B. T. M. Tauris, N. Langer, and P. Podsiadlowski. "Erratum: Ultra-stripped supernovae: progenitors and fate." en. *Monthly Notices of the Royal Astronomical Society*, Jan. 2017, stx223. ISSN: 0035-8711, 1365-2966. DOI: 10.1093/mnras/stx223. URL: https://academic.oup.com/mnras/article-lookup/doi/10.1093/mnras/stx223.

[49] T. M. Tauris et al. "Formation of Double Neutron Star Systems." *The Astrophysical Journal*, **846**(2), Sept. 2017, p. 170. ISSN: 0004-637X, 1538-4357. DOI: 10.3847/1538-4357/aa7e89. URL: https://iopscience.iop.org/article/10.3847/1538-4357/aa7e89.

[50] P. C. Peters and J. Mathews. "Gravitational Radiation from Point Masses in a Keplerian Orbit." en. *Physical Review*, **131**(1), July 1963, pp. 435–440. ISSN: 0031-899X. DOI: 10.1103/PhysRev.131.435. URL: https://link.aps.org/doi/10.1103/PhysRev.131.435.